

GDP360 – Analyzing 50 Years of Global Growth Trends

Background Story

Over the last five decades, the global economy has undergone profound structural and geopolitical transformations — from the oil shocks and industrial revolutions to globalization, digitalization, and the COVID-19 pandemic. In this evolving landscape, nations have alternated between rapid expansion and economic stagnation, shaping today's complex global order.

To help governments, analysts, and academic researchers gain a 360-degree understanding of these shifts, the **World Economic Insights Council (WEIC)** launched GDP360 — a data-driven initiative aimed at building a historical and analytical view of the world's top 20 economies between 1975 and 2025.

In this hackathon, participants will act as Data Analysts and System Designers responsible for creating a KNIME-powered analytical ecosystem that processes, transforms, and visualizes five decades of GDP and macroeconomic indicators. The final deliverable will demonstrate how KNIME Analytics Platform, in conjunction with SQL-based reasoning, can replicate the entire ETL and BI lifecycle within a single integrated environment.

Hackathon Overview

The GDP360 Hackathon is designed as a **210-minute guided challenge** that simulates the workflow of a professional data analytics project.

The session is divided into two major halves:

Part I – Conceptual (Low-Code) Segment

Participants explore the fundamentals of database design, data access, and analytical querying through SQL-oriented reasoning and visualization concepts — without writing code. This builds conceptual clarity on data architecture and economic interpretation.

Part II – KNIME Implementation Segment

Participants use the KNIME Analytics Platform to replicate all stages of the data pipeline — from ingestion to dashboard creation — using guided lab exercises. This reinforces end-to-end practical proficiency in a no-code/low-code environment.

Throughout the session, mentors guide participants step-by-step through a realistic data workflow, mirroring the processes followed by professional data engineers and analysts in enterprise environments.

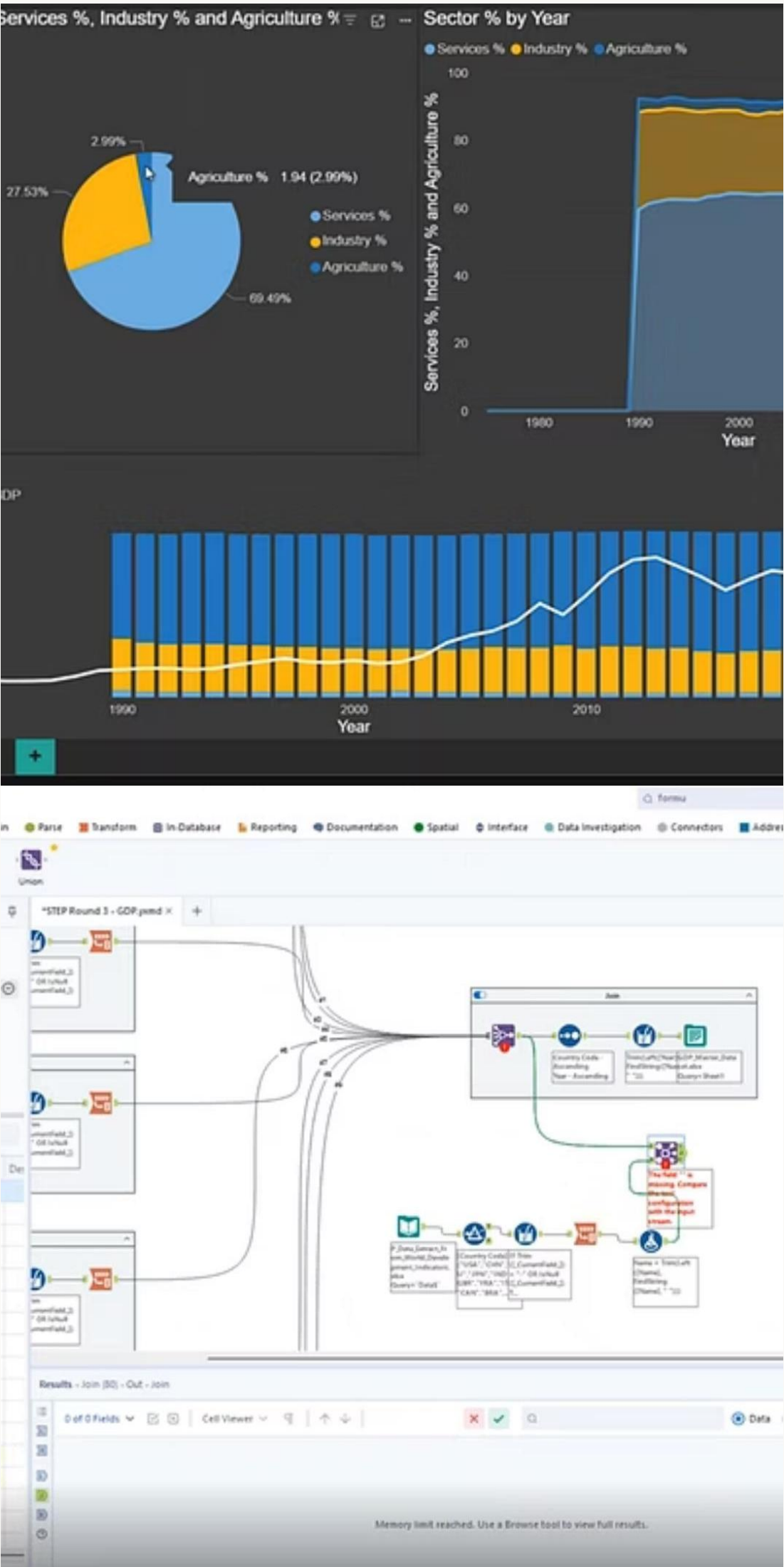
Actual Screenshots of Hackathon done by Previous Year Trainee using Alteryx F PowerBI

This document showcases the comprehensive GDP360 Hackathon experience, featuring real examples from previous participants and a detailed breakdown of the 210-minute analytical challenge that transforms raw economic data into actionable insights.

Previous Year Hackathon Screenshots

The following screenshots demonstrate the quality and depth of work achieved by previous year trainees during the hackathon using Alteryx & PowerBI. These examples showcase the level of analytical sophistication and visual storytelling expected in the GDP360 challenge.

Access the Hackathon Recording by using the following link: [Hackathon Recording](#)



Core Implementation Modules F Scenarios

1. Context Setting F End Goal Visualization

Scenario:

Participants are introduced to the GDP360 challenge, its economic relevance, and the expected outcomes. They explore the nature of global GDP trends and examine the final deliverable – interactive dashboards that illustrate historical growth, sectoral transitions, and macroeconomic dependencies.

Requirements:

- Understand the GDP360 story, objectives, and analytical vision.
- Review sample visual outputs to identify key patterns and insights.
- Establish alignment on deliverables, tools, and workflow stages.

Why:

This phase sets the strategic and analytical context, helping participants visualize the purpose and value of the tasks ahead.

Database Design F Analytical Queries

2. Database Design F Data Import (Conceptual)

Scenario:

Before working in KNIME, participants explore how GDP data would be organized within a normalized database schema. This conceptual exercise ensures clarity on how entities like Countries, Years, and Metrics interact.

Requirements:

- Understand schema design (Countries, Years, GDP_Metrics, Sectors).
- Learn to map relationships (one-to-many, many-to-one) across tables.
- Explore how SQL queries would retrieve GDP and sectoral data efficiently.

Why:

A sound conceptual model ensures that KNIME workflows mirror structured database logic, resulting in clarity and scalability.

3. Queries F Insights (Conceptual)

Scenario:

Participants simulate the process of writing analytical SQL queries without actual execution. They reason through use-cases such as ranking top economies, computing Year-over-Year (YoY) growth, and identifying dominant sectors by country.

Requirements:

- Explore logical query design for GDP ranking, growth analysis, and sectoral performance.
- Understand window functions, conditional aggregations, and grouping logic.
- Discuss expected output formats and how they translate into insights.

Why:

Developing an SQL-style analytical mindset prepares participants to perform equivalent transformations in KNIME with the same reasoning and accuracy.

KNIME Data Processing Pipeline

01	02	03
Data Access & Cleaning (KNIME)	Data Transformation & Derivation (KNIME)	Merging & Aggregation (KNIME)
Scenario: The raw GDP dataset from the World Bank is loaded into KNIME for initial exploration and cleaning. Participants encounter typical data inconsistencies such as null values, inconsistent naming, and missing years.	Scenario: Cleaned datasets are transformed into analytically rich forms. Participants compute derived indicators like GDP per capita, CAGR, and YoY growth, while restructuring data for dimensional slicing.	Scenario: To understand macroeconomic interdependencies, participants integrate GDP data with other indicators such as Population, Inflation, and Debt. The combined dataset is analyzed to derive correlations and country-level aggregates.
Requirements: Use KNIME File Reader / Excel Reader to load datasets. Apply nodes like String Manipulation, Missing Value, Rule Engine, and GroupBy. Normalize country names, fix schema inconsistencies, and flag anomalies.	Requirements: Implement transformations using Math Formula, Lag Column, GroupBy, and Window nodes. Reshape data to normalized structures suitable for analysis. Validate metrics using calculated benchmarks and sanity checks.	Requirements: Merge multiple datasets using Joiner and Concatenate nodes. Calculate regional aggregates and time-based summaries. Identify key macroeconomic drivers of GDP growth.
Why: Data cleaning ensures reliability and analytical accuracy — a fundamental prerequisite for any subsequent transformation or visualization.	Why: Transformation bridges raw data and insights — converting static information into dynamic economic narratives.	Why: This module demonstrates the analytical depth possible within KNIME by merging multi-dimensional datasets to draw policy-level insights.

Visualization F Final Deliverables

7. Visualization F Exporting (KNIME)

Scenario:

Participants create final dashboards within KNIME to visualize GDP patterns, sectoral compositions, and comparative growth metrics. Visual outputs are then exported for reporting.

Requirements:

- Use visualization nodes (Line Plot, Bar Chart, Heatmap, Pie Chart).
- Incorporate interactive filters (country, year).
- Export dashboards and reports as PDFs using KNIME reporting extensions.

Why:

Data storytelling transforms analytical outputs into actionable insights — an essential skill in presenting complex information to decision-makers.

8. Wrap-Up F Insight Discussion

Scenario:

The session concludes with a discussion on analytical findings, economic interpretations, and reflections on the KNIME workflow experience.

Requirements:

- Interpret visualization results and share key takeaways.
- Discuss how macro indicators influenced GDP outcomes.
- Reflect on the efficiency of using KNIME as a unified ETL and BI platform.

Why:

This final stage consolidates learning, emphasizes insight generation, and highlights the professional applications of KNIME in real-world analytics.

Detailed Hackathon Schedule

The GDP360 Hackathon follows a carefully structured **210-minute timeline** designed to balance conceptual learning with hands-on implementation. Below is the complete phase-by-phase breakdown:

Phase	Time (min)	Start Time	End Time	Description
Context Setting & End Goal Visualization	30	9:00	9:30	Introduce the GDP360 Hackathon, explain objectives, and showcase the final expected dashboards or visual outputs to set the vision.
Database Design & Data Import	20	9:30	9:50	Demonstrate how the GDP data is structured conceptually; explain schema, country-year relationships, and import sample datasets.
Queries & Insights	30	9:50	10:20	Discuss SQL-style analytical thinking — how queries extract insights like growth rate, rankings, and trends using logic rather than execution.
Data Access & Cleaning	30	10:20	10:50	Explain how raw data from the World can be accessed and cleaned — handling missing values, inconsistent names, and anomalies.
Break	20	10:50	11:10	Short break before moving into KNIME-based practical implementation.
Data Transformation & Derivation	30	11:10	11:40	Hands-on guided lab in KNIME to calculate derived metrics like YoY Growth, GDP per capita, and CAGR using transformation nodes.
Merging & Aggregation	30	11:40	12:10	Guided lab where participants merge GDP with macro indicators (Inflation, Population, Debt) and perform aggregations and correlations in KNIME.
Visualization & Exporting	30	12:10	12:40	Create dashboards and charts within KNIME (line charts, bar charts, heatmaps) and export the final report as PDF.
Wrap-Up & Insight Discussion	10	12:40	12:50	Discuss findings, summarize learning outcomes, and reflect on key insights derived from the GDP analysis.
Attendance 2	10	12:50	1:00	Final attendance, feedback collection, and closure of the hackathon session.

Key Takeaways F Learning Outcomes

The GDP360 Hackathon provides participants with a comprehensive, hands-on experience that mirrors real-world data analytics workflows used by professional data engineers and business intelligence specialists.

1

End-to-End Data Pipeline Mastery

Participants gain practical experience in every stage of the analytics lifecycle — from raw data ingestion and cleaning to transformation, aggregation, and visualization — all within a unified KNIME environment.

2

SQL-Style Analytical Thinking

By developing conceptual understanding of database design and query logic before implementation, participants build a strong foundation for structured analytical reasoning that translates across platforms.

3

Economic Insight Generation

Beyond technical skills, participants learn to interpret macroeconomic patterns, understand the drivers of GDP growth, and communicate complex findings through compelling data visualizations.

4

Professional Workflow Simulation

The hackathon replicates enterprise-level analytics projects, preparing participants for real-world challenges in data engineering, business intelligence, and economic research roles.

 **Ready to Transform Data into Insights?** The GDP360 Hackathon demonstrates that with the right tools, structured methodology, and analytical mindset, anyone can unlock the stories hidden within complex economic datasets and drive meaningful decision-making.